

Pengenalan Tools dan Custom Tools pada CrewAI

Pada crewai, kita diberikan berbagai macam tools untuk menunjang pekerjaan dari Agent.

Secara detail dapat dilihat pada link berikut:

<https://docs.crewai.com/en/concepts/tools>

Contoh kasus penggunaan tools

Misal kita ingin membuat agent yang dapat membaca sebuah document (text/txt => karena ringan) dan memberikan report-report penting, kita dapat awali dengan membuat sebuah crew dengan perintah:

Crewai flow add-crew crew_rag

Maka akan secara otomatis menambahkan sebuah folder yang ada didalam folder crews dengan nama [crew_rag]. Setelah itu, kita ganti agents dan tasknya masing-masing menjadi:

Agents.yaml

```
1 doc_analyzer:
2   role: >
3     document extractor for {file}
4   goal: >
5     extract from {file}
6   backstory: >
7     You are meticulous when reading documents, so you can report your extraction result
8
9
10
```

Tasks.yaml

irc > Indoprimaflow > crews > crew_rag > config > ! tasks.yaml

```
1 documen_analyzer_task:
2   description: >
3     Conduct a thorough extract of an existing {file}, and make sure you identify
4     every part from the report, particularly those related to sales figures.
5   expected_output: >
6     A list with 5 insight about the document
7   agent: doc_analyzer
8
9
```

Setelah itu, kita perlu menambahkan salah satu tool (FileReadTool) pada file crew_rag.py dengan cara import:

```
c > indoprimaflow > crews > crew_rag > crew_rag.py > ...
1  from crewai import Agent, Crew, Process, Task
2  from crewai.project import CrewBase, agent, crew, task
3  from crewai.agents.agent_builder.base_agent import BaseAgent
4  from crewai_tools import RagTool, PDFSearchTool, FileReadTool
5  # If you want to run a snippet of code before or after the crew starts,
6  # you can use the @before_kickoff and @after_kickoff decorators
7  # https://docs.crewai.com/concepts/crews#example-crew-class-with-decorators
8
```

Pada tutorial kali ini, kita tidak akan menggunakan ragtool ataupun pdfsearchtool. Kita hanya fokus pada FileReadTool yang paling ringan untuk segi pemrosesan komputasi.

Lalu, kita bisa memanggil tools dengan cara:

```
fileRead = FileReadTool()
```

Setelah itu, kita bisa setting agents dan task seperti berikut:

```

@CrewBase
class CrewRag():
    """CrewRag crew"""

    agents: list[BaseAgent]
    tasks: list[Task]

    agents_config = "config/agents.yaml"
    tasks_config = "config/tasks.yaml"

    fileRead = FileReadTool()

    @agent
    def doc_analyzer(self) -> Agent:
        return Agent(
            config=self.agents_config['doc_analyzer'], # type: ignore[index]
            verbose=True,
            tools=[self.fileRead]
        )

    @task
    def research_task(self) -> Task:
        return Task(
            config=self.tasks_config['documen_analyzer_task'], # type: ignore[index]
        )

```

Terlihat pada bagian agent, kita tambahkan 1 parameter bernama tools. Dimana parameter tersebut berfungsi untuk membaca file input. Karena pada bagian agents dan tasks.yaml kita sudah mendefinisikan file dengan nama {file}. Maka untuk celery task kita bisa membuat dengan cara berikut:

```

@celery_app.task(bind=True, name="doc_analyzer")
def doc_analyzer(self, file:str):
    self.update_state(state='RUNNING', meta={'current':f'start job for{file}'})
    try:
        result = CrewRag().crew().kickoff(inputs = {"file": file})
        return str(result)
    except Exception as e:
        # self.update_state(state='FAILURE', meta={'error':str(e)})
        logger.error(f"Task failed with error: {e}\n{traceback.format_exc()}")
        raise

```

Dimana def doc_analyzer() memiliki input berupa variable file: str (menyesuaikan {file} yang ada pada YAML).

Setelah itu, kita dapat membuat REST APInya dengan cara:

```
@app.post("/rag")
async def rag(file: UploadFile = File(...)):
    if file.content_type != "text/plain": # Use direct comparison, not startswith
        raise HTTPException(status_code=400, detail="File must be a PDF")

    # Save the uploaded file to a temporary location
    # Generate a unique filename to avoid collisions
    file_extension = os.path.splitext(file.filename)[1] or ".pdf" # Changed from .jpg
    unique_filename = f"{uuid.uuid4().hex}{file_extension}"
    file_path = os.path.join(FILE_FOLDER, unique_filename)

    # Save the file
    content = await file.read()
    with open(file_path, "wb") as f:
        f.write(content)

    # Make sure 'task' is defined before returning

    task = celeryTask.doc_analyzer.delay(file_path)
    return {"task_id": task.id, "file_path": file_path}
```

Dimana kita terlebih dahulu melakukan checking terhadap file yang di upload, pada case ini kita pastikan file yang diupload adalah text/txt saja. Karena tool yang kita gunakan hanya mampu untuk membaca file txt (untuk mendemokan proses yang ringan saja).

Untuk mengirim file, kita bisa gunakan postman, dengan cara:

The screenshot shows a Postman interface for a new request. The method is POST and the URL is http://127.0.0.1:8000/rag. The 'Body' tab is selected, and the 'form-data' radio button is chosen. A single key-value pair is added: 'file' with a file named 'MONTHLY GRAPE SALES REPORT.txt'. The response is shown in the 'Body' tab at the bottom, indicating a 200 OK status. The JSON response contains a task_id and a file_path.

Key	Value	Description
file	File MONTHLY GRAPE SALES REPORT.txt	

Key	Value	Description
task_id	fe5487bc-f326-4bff-a5ac-113fc48265fa	
file_path	/var/home/gustipang/Documents/project/indoprimalflow/file/14c569f5ac5e40ee9fbc29c384d49b99.txt	

Sesuai dengan nilai return yang sudah kita set, kita mendapatkan 2 nilai, task_id dan juga file_path. File_path merupakan nama file yang telah kita kirimkan dan akan disimpan ke dalam file project kita.

Lalu, hasil dari proses task_id bisa kita lihat dengan cara menggunakan HTTP request /status:

HTTP

indoprima_RAG > check task

Save

Share

GET

http://127.0.0.1:8000/status/fe5487bc-f326-4bff-a5ac-113fc48265fa

Send

Docs

Params

Authorization

Headers 7

Body

Scripts

Settings

Co

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body

Cookies

Headers 4

Test Results

200 OK

4 ms

2.09 KB

Save Response

{ } JSON

Preview

Visualize

```
1 {
2   "task_id": "fe5487bc-f326-4bff-a5ac-113fc48265fa",
3   "status": "SUCCESS",
4   "result": "MONTHLY GRAPE SALES REPORT - ARAB MARKETS\n=====
\nReporting Period: Q1 2026 (January - March)
\nMarkets Covered: UAE, Saudi Arabia, Egypt, Jordan, Lebanon\n\nSALES BY COUNTRY (USD Millions)\n-----|-----|
-----|-----|-----|-----\nCountry      | Jan   | Feb   | Mar   | Total | Growth%\n-----|-----|-----|-----|-----|-----|
\nUAE          | 1.45  | 1.62  | 1.89  | 4.96  | +18.2%\nSaudi Arabia | 1.85  |
2.01  | 2.34  | 6.20  | +22.5%\nEgypt       | 0.52  | 0.48  | 0.61  | 1.61  | +8.3%\nJordan      | 0.31  | 0.
35   | 0.42  | 1.08  | +12.9%\nLebanon     | 0.18  | 0.22  | 0.27  | 0.67  | +15.0%\n\nGRAPE VARIETIES
PERFORMANCE\nVariety      | Market Share | Avg Price/kg | Customer Rating\n-----|-----|-----|-----|
\nRed Globe      | 42%          | $6.50        | 4.8/5\nThompson Seedless |
35%          | $5.20        | 4.6/5\nBlack Magic      | 15%          | $7.80        | 4.9/5\nFlame Seedless   |
8%          | $6.90        | 4.5/5\n\nKEY INSIGHTS\n- Ramadan Effect: March sales increased by 28% due to Ramadan 2026\n-
Premium Segment: Black Magic grapes saw 45% volume growth in UAE\n- Export Trend: Increased demand for seedless varieties (+32%
YoY)\n- Seasonal Dip: Lower sales in Egypt due to domestic harvest season\n\nRECOMMENDATIONS\n1. Increase Red Globe inventory by
40% for April-May period\n2. Launch marketing campaign for Black Magic in Saudi Arabia\n3. Consider price optimization for
Thompson Seedless\n4. Expand cold storage capacity in Dubai logistics hub\n\nDATA QUALITY NOTES\n- Data source: Wholesale market
surveys and distributor reports\n- Exchange rates: USD stable across reporting period\n- Missing data: Oman and Qatar (not
included this quarter)\n- Timeliness: Data verified as of April 15, 2026",
5   "error": null
6 }
```

Untuk result terlihat panjang, dikarenakan pada bagian task kita meminta agent untuk melakukan analisa. Kita juga bisa membuat task untuk menampilkan point-point atau bahkan result berupa JSON.

Menambahkan Agent untuk mengolah data yang sudah diolah oleh Agent sebelumnya

Hasil dari agent sebelumnya sepertinya masih kurang begitu mudah dipahami. Untuk itu, kita perlu membuat agent lebih fokus kepada penyampaian data yang lebih simple. Cara paling mudah yang dapat dilakukan adalah kita menambahkan 1 agent lagi untuk mengolah hasil dari agent sebelumnya (doc_analyzer agent).

Pada tahap ini, kita akan membuat 1 agent lagi dengan nama [doc_resumer], dimana agent ini berfungsi untuk melakukan resume terhadap hasil analisa dari agent doc_analyzer. Untuk membuat agent tersebut, kita dapat menambahkan agent pada file agents.yaml:

Agents.yaml

```
1 doc_analyzer:
2   role: >
3     document extractor for {file}
4   goal: >
5     extract from {file}
6   backstory: >
7     You are meticulous when reading documents, so you can report your extraction result
8
9
10 doc_resumer:
11   role: >
12     Senior Report Simplification Specialist
13   goal: >
14     Transform complex and lengthy reports into exactly 5 clear,
15     easy-to-understand, and actionable insight points
16   backstory: >
17     You are a data communication expert with 15 years of experience
18     simplifying complex business reports. You have a unique ability
19     to cut through information noise and extract the core essence
20     from any report. Your clients praise your ability to make
21     complicated data understandable for all management levels.
22     Your principle: "If you can't explain it in 5 simple points,
23     you haven't truly understood it yet."
```

Kita setting role, goal, dan juga backstorynya sedemikian rupa sehingga agent mengenali dirinya sebagai seorang senior report simplification specialist. Lalu, kita atur sedemikian rupa pada bagian tasks.yaml agar outputnya lebih terarah dan sesuai dengan yang kita harapkan:

Tasks.yaml

```

doc_resumer_task:
description: >
  INPUT ANALYSIS:
  The original report to be simplified is as follows:

  SPECIFIC INSTRUCTIONS:
  Your task is to analyze the above report and generate exactly
  5 (FIVE) insight points that are:

  1. **CLEAR**: Use everyday language that is easy to understand
  2. **CONCISE**: Maximum 2 sentences per point
  3. **IMPACTFUL**: Highlight the most important/interesting findings
  4. **STRUCTURED**: Each point contains one main idea
  5. **ACTIONABLE**: Provides clear direction or recommendations

  EXPECTED OUTPUT FORMAT:

  # 5 KEY INSIGHTS: [Brief Report Title]

  ## Insight 1: [Insight Title]
  [1-2 sentence clear and concise explanation]

  ## Insight 2: [Insight Title]
  [1-2 sentence clear and concise explanation]

  ## 4 Insight 3: [Insight Title]
  [1-2 sentence clear and concise explanation]

  ## Insight 4: [Insight Title]
  [1-2 sentence clear and concise explanation]

  ## Insight 5: [Insight Title]
  [1-2 sentence clear and concise explanation]

  ---

  WORKFLOW:
  1. Read and understand the entire report
  2. Identify the 5 most critical findings
  3. Rewrite each finding in simple language
  4. Add relevant icons for visual appeal
  5. Review whether each point meets the above criteria

  ---

  ---

  IMPORTANT RULES:
  - DO NOT generate more than 5 points
  - DO NOT use technical jargon without explanation
  - DO NOT copy sentences directly from the original report
  - MUST use proper English grammar
  - MUST include a title for each insight

  expected_output: >
  5 structured insight points in the specified format,
  each with a title and 1-2 sentence explanation.
  Total output no more than 15 lines for explanations.
agent: doc_resumer
context:
  - documen_analyzer_task

```

Tidak lupa kita tambahkan [context] pada bagian akhir task, fungsi dari context adalah untuk membuat task yang dijalankan agent akan berfokus kepada hasil yang dihasilkan

oleh agent yang dideskripsikan di dalam context, dalam kasus ini kita tambahkan task dari agent doc_analyzer, yaitu [documen_analyzer_task].

Setelah itu, tidak lupa kita buat juga class @agent dan @task pada file crew_rag.py:

```
@agent
def doc_analyzer(self) -> Agent:
    return Agent(
        config=self.agents_config['doc_analyzer'], # type: ignore[index]
        verbose=True,
        tools=[self.fileRead]
    )

@agent
def doc_resumer(self) -> Agent:
    return Agent(
        config= self.agents_config['doc_resumer'],
        verbose= True,
    )

@task
def documen_analyzer_task(self) -> Task:
    return Task(
        config=self.tasks_config['documen_analyzer_task'], # type: ignore[index]
    )

@task
def doc_resumer_task(self) -> Task:
    return Task(
        config= self.tasks_config['doc_resumer_task'],
    )
```


Dengan menggunakan 2 agent yang saling bekerja sama, kita telah membuat sebuah crew yang saling bekerja sama untuk mengerjakan sebuah task, outputnya seperti ini:

GET

http://127.0.0.1:8000/status/2032a616-5500-490a-b284-d0c8d2f50dca

Send

Docs

Params

Authorization

Headers 7

Body

Scripts

Settings

Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body

Cookies

Headers 4

Test Results

200 OK • 5 ms • 1.59 KB • Save Response

JSON

Preview

Visualize

```
1 {
2   "task_id": "2032a616-5500-490a-b284-d0c8d2f50dca",
3   "status": "SUCCESS",
4   "result": "# 📊 5 KEY INSIGHTS: Q1 2026 Arab Market Grape Performance\n## 🌐 Insight 1: Regional Growth Leaders\nSaudi Arabia and the UAE showed the strongest sales growth, indicating these markets are the primary drivers for expansion. Focus marketing and inventory efforts on these two countries to maximize overall regional sales.\n## 💡 Insight 2: Ramadan's Sales Impact\nThe sales in March saw a significant jump, especially driven by the Ramadan effect, which suggests seasonal events strongly influence purchasing patterns. Plan inventory accordingly for peak seasonal demand in upcoming months.\n## ✂️ Insight 3: Premium Product Opportunity\nThe high-end 'Black Magic' variety experienced substantial volume growth in the UAE, pointing to an untapped opportunity in the premium segment. Launch targeted campaigns to capitalize on this high-value customer interest.\n## 📈 Insight 4: Export and Variety Shift Trend\nThere is a clear market shift towards seedless grape varieties, with demand increasing by over 32% year-over-year. Prioritize expanding the supply chain and marketing efforts for these convenient options.\n## 🌱 Insight 5: Address Regional Disparities\nSales in Egypt experienced a noticeable dip, likely due to domestic harvest cycles, while other markets remain stable. Adjust sourcing and promotional strategies to account for these regional seasonal differences moving forward.",
5   "error": null
6 }
```